Supplementary Preregistration: Complexity and Creativity in Human–LLM Interactio (#300809)

Author(s)
This pre-registration is currently anonymous to enable blind peer-review.
It has 2 authors.

Pre-registered on:
2026/07/08 05:52 (PT)

1) Have any data been collected for this study already?
It's complicated. We have already collected some data but explain in Question 8 why readers may consider this a valid pre-registration nevertheless.

2) What's the main question being asked or hypothesis being tested in this study?
This preregistration examines whether users' prompts to GPT vary in complexity and creativity, and whether GPT-generated responses vary in creativity, as a function of age group and task context. Specifically, we test whether prompt complexity and creativity, as well as GPT response creativity, differ by age group, task context, and their interaction, and whether prompt-level measures change across successive turns within the interaction.
The tasks differed in their interaction demands: the practice task served as an initial open-ended interaction; the trip-planning task emphasized information-seeking, planning, and specifying constraints; and the greeting task emphasized expressive writing and creative formulation. We therefore examine whether these task demands are reflected in users' prompts and GPT response creativity, and whether these patterns differ by age group.
Differences in prompt complexity and creativity may reveal how users from different age groups formulate instructions for GPT and translate task goals into prompts. Differences in response creativity may further indicate whether GPT-generated outputs vary across users' prompts, task contexts, or age groups. These analyses are preregistered before conducting the relevant semantic analyses of prompt complexity and creativity.

3) Describe the key dependent variable(s) specifying how they will be measured.
The primary dependent variables in this preregistration are:
Prompt complexity will be measured at the message level using PrediQT (Raz et al., 2026), a pretrained model that estimates question complexity. Although participants' inputs to GPT are not always phrased as explicit questions, they function as prompts or requests within the interaction. Therefore, PrediQT will be applied separately to each user message sent to GPT, producing a continuous complexity score for each prompt/message.
Prompt creativity and Response creativity will be operationalized as two separate measures using the Divergent Semantic Integration (DSI) score (Johnson et al., 2022). DSI is an embedding-based measure that captures the extent to which a text integrates semantically distant concepts into a coherent text. Prompt creativity will be computed for each user prompt/message sent to GPT, whereas response creativity will be computed for each GPT-generated response. Higher DSI scores indicate greater semantic integration and higher semantic creativity.
In addition, variables already collected in the original study will be used, including the number of messages per interaction, message length/word count, participants' self-ratings of GPT's responses, and external evaluations of GPT's responses.

4) How many and which conditions will participants be assigned to?
Participants were divided into two age groups: younger and older adults. All participants completed three GPT interaction tasks: a practice task, a greeting task, and a trip task. Thus, age group is a between-participants factor, and task is a within-participants factor.
Most analyses in this preregistration will include all three tasks when the relevant prompt-level measures are available. Analyses involving participants' self-evaluations of GPT's responses will include only the greeting and trip tasks, because these ratings were not collected for the practice task.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.
Prompt complexity and prompt creativity will be analyzed using linear mixed-effects models at the message level. Scores will be predicted from task, age group, and their interaction, with the participant included as a random intercept. Additional covariates will include message length/word count, message order within the interaction, and the total number of messages in the interaction.
The main models will be:
Prompt complexity ~ Task × Age Group + word count + message order + total number of messages + (1 | participant)
Prompt creativity ~ Task × Age Group + word count + message order + total number of messages + (1 | participant)
Follow-up comparisons will be conducted for significant Task effects or Task × Age Group interactions, with appropriate correction for multiple comparisons. Significant Age group effects will be interpreted based on the model coefficients.
In additional analyses, self-evaluations of GPT's responses will be examined as dependent variables at the interaction level. Linear mixed-effects models will be used to predict self-evaluation scores from average prompt complexity, average prompt creativity, average GPT response creativity, Age group, Task type, and their relevant interactions. Additional covariates will include interaction-level variables, such as the total number of messages in the interaction, with the participant included as a random intercept.
GPT response creativity will be examined in a separate set of analyses. Linear mixed-effects models will be used to predict GPT response creativity from prompt complexity, prompt creativity, Age group, Task type, and their relevant interactions. Additional covariates will include prompt word count, GPT response word count, message order within the interaction, and the total number of messages in the interaction, with the participant included as a random intercept.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.
No additional participant-level exclusions will be applied beyond those defined in the original study. At the message level, prompts will be excluded only if they are missing, empty, non-textual, or not processed by the relevant computational model (e.g., due to a minimum text length

requirement).

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

The present preregistration is based on an existing dataset collected as part of the original study. The sample includes 90 participants: 45 younger adults and 45 older adults. Each participant completed three GPT interaction tasks: a practice task, a greeting task, and a trip task.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

This preregistration concerns additional analyses of an existing dataset. Although the data have already been collected, the present prompt-level semantic analyses were not specified in the original preregistration or analysis plan and will be conducted only after this preregistration is submitted. The goal of these analyses is to examine additional aspects of human–GPT interaction that were not included in the original analysis plan, particularly whether prompt complexity, prompt creativity, and response creativity vary across task contexts and age groups.

As secondary or exploratory analyses, we may examine embedding-based semantic similarity or distance between users' prompts, between users' prompts and the original task instructions, and across prompts within the same task. We may also examine associations between different prompt-level, response-level, and evaluation-based measures collected or computed in the study.